

MODEL QUESTION PAPER

MATERIAL SCIENCE AND TECHNOLOGY

*Time : 3 Hour**Max.Marks : 75***PART A**I. Answer **all** questions in one word or one sentence**(9x 1 = 9 Marks)**

- 1 Define stress
- 2 The internal components of the material is usually related by its
- 3 The stress level at which plastic deformation begins is known as
- 4 Give an example of piezoelectric material.
- 5 Define diamagnetism
- 6 Steel is a alloy
- 7 Identify the structure of diamond
- 8 Name a thermosetting polymer
- 9 Define polymer matrix composite

PART BII. Answer any **eight** questions from the following. Each carries 3 marks **(8 x 3 = 24 Marks)**

- 1 List the four components of material science and engineering and their inter relationship.
- 2 List basic three classification of solid materials with example
- 3 Define unit cell.
- 4 Differentiate between edge dislocation and screw dislocation
- 5 Illustrate the stress strain diagram of elastic deformation for loading and unloading cycle.
- 6 Define corrosion penetration rate.
- 7 Identify a suitable alloy for the manufacture of aircraft body parts. Enumerate any two property of the alloy chosen.
- 8 List any three application of polycarbonates
- 9 Distinguish between thermoplastic and thermosetting polymer
- 10 Sketch the schematic diagram for pultrusion process

PART C

Answer all questions. Each question carries 7 marks

(6 x 7 = 42 Marks)

III Discuss the types of polymers with examples

OR

IV Illustrate the working of diffractometer

V Define

- a) Poisson's ratio
- b) Tensile strength
- c) Ductility

OR

VI Illustrate the energy band structures in solids

VII Illustrate the tension test for a tensile sample.

OR

VIII Illustrate the magnetic hysteresis for ferromagnetic materials

IX Describe various forming operation.

OR

X Describe various methods for precipitation hardening.

XI Classify various forms of corrosion. Also define each form of corrosion

OR

XII Discuss the classification of steel.

XIII Illustrate the glass forming technique

OR

XIV Describe the types of refractories with its applications.